Cog Works - Technical Sheet

Tech

One self-contained HTML file (about 19 KB). HTML5 Canvas 2D with devicePixelRatio scaling up to 2x. Pointer Events for touch, mouse and pen. Web Audio API for all sound and music, generated in the browser. localStorage for saves. No libraries, images or network requests.

Systems

Simulation: fixed 20 Hz timestep, separate from rendering, so the game runs the same on slow and fast phones.

Logistics: each tile holds one building. Miners, belts and machines hand items to the tile they face. Belts carry one item and animate it moving across the tile.

Recipes: smelters (ore to plate) and assemblers (gear, circuit) with input buffers of 6 per ingredient. Circuits are picked first when both recipes are possible.

Economy and goals: 6 goals in a row. Goal N pays N x $100. Income per second is a rolling 10-second average.

Graphics: neon glow (canvas shadows), animated chevrons on belts, pulsing machines, a spinning assembler, smelter smoke, ore sparkle, particle bursts, floating cash, screenshake and a starfield.

Audio: synth sounds for placing, errors, sales, removing and goals. A 104 BPM synth loop (saw bass with filter sweep, kick, noise hats) whose filter opens up as you clear goals.

Audit 1 - Balance (simulated play)

A headless bot built factories on 20 random maps for 30 game minutes each. **Found:** starting cash ($120) was below the cost of the first working line ($130), so no bot ever got started. **Fixed:** starting cash raised to $150. After the fix every bot cleared Goal 1 (about 18 s on average).

Found: Goal 2 (15 copper plates) took too long on maps where copper was hard to reach. **Fixed:** lowered to 10. The average time to Goal 2 dropped from 117 s to about 70 s.

Bots that built a circuit line cleared Goal 4 in about 4 minutes and all 6 goals in about 10 minutes. The simple bot could not always find room for a circuit line on every map, but a human can route belts freely.

Audit 2 - Bugs

A scripted run in a mock browser: new game, building a full line, rotating, removing, an invalid miner, tapping off the grid, 60 s of play, save and reload, and keyboard input. No errors. The line sold 35 plates and cleared Goal 1.

Fixed while building: belt drags that went off the grid or jumped diagonally could set the wrong direction, so now only neighbouring tiles connect. Audio starts on the first tap, which iPhones need, and the game still works if Web Audio is not available.

Audit 3 - Player experience

Added a first-run tip that explains the starter line. Clear error messages for an occupied tile, a miner off ore and not enough cash, with a buzz and screenshake. A progress bar on the current goal. Big 58 px buttons in a two-row bar, safe-area padding for notched phones. Help pauses the game. 50% refund on Remove so mistakes are cheap.

Known limit: items in transit and in machine buffers are not saved, only the buildings. Tested with simulated play, not on a real phone.